Day 3

Configuring Basic Network HardwareTheory (30 min): IP addressing basics and network configuration.Practical (30 min):Set up a router with default settings

TASK – 1

Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page.

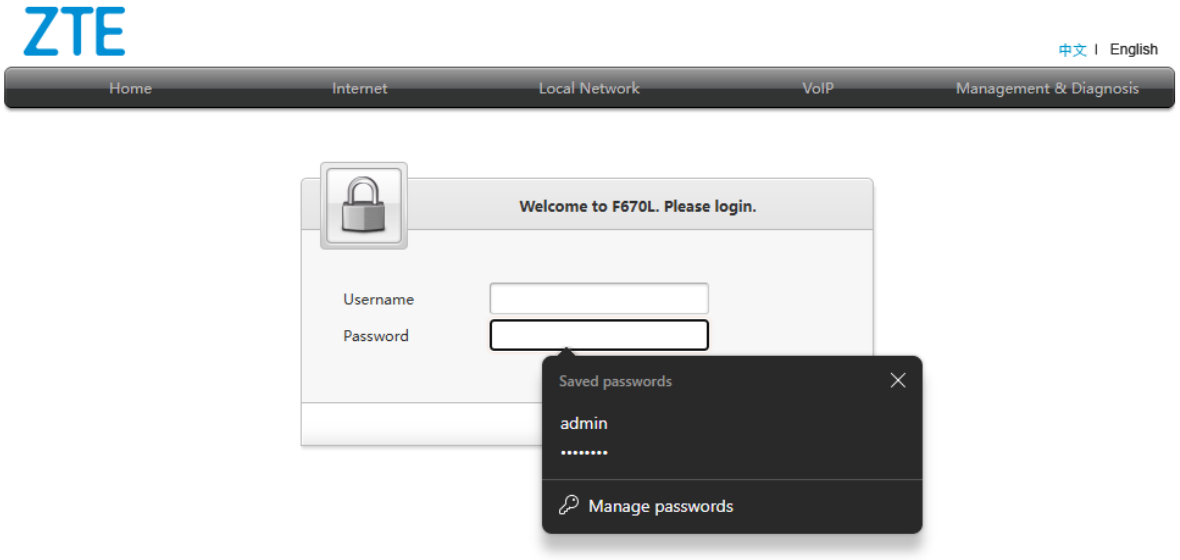
Hint: If you don't know your router's IP, open Command Prompt and run 'ipconfig' — look for 'Default Gateway'.

➔

Aim: To access the router's admin panel using its default gateway IP address.

Procedure:

1. Connect your laptop to your home Wi-Fi.
2. Press `Windows + R`, type `cmd`, and press Enter.
3. Type `ipconfig` and press Enter.
4. Find Default Gateway under the Wi-Fi adapter.
5. Open Chrome or Edge and enter that IP address in the address bar.
6. The router login page should appear. Enter the admin credentials provided on the router label or by your ISP.



TASK – 2

Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.

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Wireless LAN adapter Wi-Fi :

Connection-specific DNS Suffix :

IPv6 Address. . . . . : 2402:a00:172:81ae:817e:24c9:8cbb:92b9

Temporary IPv6 Address. . . . . : 2402:a00:172:81ae:ec3b:9129:8505:3b5

Link-local IPv6 Address . . . . . : fe80::410c:a0f3:7197:69af%7

IPv4 Address. . . . . : 192.168.1.9

Subnet Mask . . . . . : 255.255.255.0

Default Gateway . . . . . : fe80::1%7

192.168.1.1

TASK – 3

Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access.

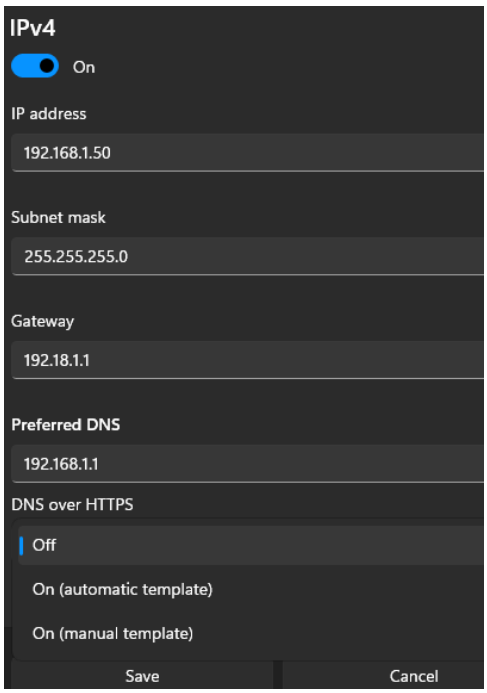
Constraint: Remember to revert to automatic/DHCP after testing to avoid network issues.

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Aim: To configure a manual IPv4 address and test the internet connection.

Procedure on Windows 11:

1. Open Settings → Network & internet → Wi-Fi.
2. Open your connected Wi-Fi network's properties.
3. Find IP assignment and click Edit.
4. Select Manual, enable IPv4, and enter the IP address, subnet prefix length, gateway, and DNS settings.
5. Save the settings and test your connection by opening a website.



In this task, I changed my laptop's IP configuration from automatic (DHCP) to a static IP address. I entered the IP address as `192.168.1.50`, the subnet mask as `255.255.255.0`, the default gateway as `192.168.1.1`, and the preferred DNS as `192.168.1.1`. After saving the settings, I tested the internet connection to check whether the configuration was working correctly.

This practical helped me understand the difference between DHCP and a static IP address. DHCP automatically assigns IP addresses to devices, whereas a static IP address is configured manually. After completing the test, I restored the settings to Automatic (DHCP) to avoid future network configuration issues

TASK – 4

In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

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Procedure

1. Open your router's admin panel using its gateway IP address, usually `192.168.1.1`.
2. Log in and look for LAN Settings, Local Network, or DHCP Server.
3. Note the DHCP Start IP Address and DHCP End IP Address.
4. DHCP Start IP Address
5. 192 . 168 . 1 . 2
6. DHCP End IP Address
7. 192 . 168 . 1 . 254



What happens if two devices have the same IP address?

If two devices on the same local network are assigned the same IP address, an IP address conflict occurs. This can cause unstable connectivity or prevent devices from communicating properly. Using DHCP correctly and avoiding duplicate static IP addresses helps prevent this problem.